Parameter Registry — Knotted Coil Designer

Every number that shapes a result, with its true origin. Tags follow the research-integrity code standard: **LIT** (from verified literature) · **CONV** (convention/library default) · **JUDG** (our judgement) · **TUNED** (searched) · **DERIVED** (computed from others). **PLACEHOLDER** marks a JUDG value awaiting Reza's real specification.

Last updated 2026-10-23.

Physical inputs (placeholders until Reza supplies them)

Parameter	Value	Tag	Rationale / source	Where
Wire diameter d_w	1.0 mm	JUDG PLACEHOLDER	Round number for magnet-wire-scale loops. Awaiting spec (Design D5)	app, tests
Insulation thickness t_i	0.25 mm	JUDG PLACEHOLDER	Awaiting spec (D5)	app, tests
Minimum bend radius	2 mm	JUDG PLACEHOLDER	Awaiting spec (D5)	app, tests
Helmet semi-axes a, b, c	85, 105, 110 mm	JUDG PLACEHOLDER	Rough adult head plus clearance, not measured. Awaiting former dimensions (D4)	`core/surface.js`
Lowest covered height zCut	~~0 mm~~ **superseded: −20 mm since v1.6** (see v1.6 table)	JUDG PLACEHOLDER	Ellipsoid equator (v1.0–v1.5)	`core/surface.js`
Face opening half-angle / top	40° / 60 mm	JUDG PLACEHOLDER	Visual guess	`core/surface.js`

Design defaults (Design.md v1.0)

Parameter	Value	Tag	Rationale / source	Where
Bloom shape	outward (lobes extend from a fixed base circle)	JUDG	Design v1.2 (Reza 2026-10-20); ‘symmetric’ (v1.0) selectable	knot, app
Petal sharpness m	1 (range 1–16)	JUDG	Design v1.3 (Reza 2026-10-20). Profile exponent; 1 = v1.2 shape. Range upper bound is judgement: bends tighten quickly beyond about 10	knot, app
Lobe aspect-ratio definition	length ÷ width at half length	DERIVED	Claude’s definition; formula in `core/knot.js` `lobeAspect`	knot, app
Bloom range β	0.02–3.0 outward, 0.02–0.9 symmetric (UI allows 0.005–3)	JUDG	Design §3.3. **Validation run 1 showed 0.02 is below the buildable floor for the placeholder wire**. See DECISIONS.md	knot, app
Decoupling threshold |k|	0.01	JUDG PLACEHOLDER	Design D6; not yet justified from literature	`core/optimise.js`
Neighbour definition	nearest ring (d < 1.35 × pair-null spacing)	JUDG	Design v1.1 §5.1 (Reza, 2026-10-20). The v1.0 rule (footprints overlap, d < 2 Rf) is selectable and always reported	`core/optimise.js`
Bloom mode	constant core	JUDG	Design D2	app
Rotation mode	free per element	JUDG	Reza 2026-10-20: rotation as a decoupling lever	`core/optimise.js`

Design v1.7: full-wave co-simulation (`sim/`)

Parameter	Value	Tag	Rationale / source	Where
Solver	openEMS 0.37.0 (FDTD), Python 3.14 wheels	CONV	Verified install 2026-10-20; docs.openems.de	`sim/run_openems.py`
Wire primitive	`AddCurve` (thin PEC curve)	JUDG	`AddWire` with radius 0.5 mm **disconnected** on a 2 mm grid (see DECISIONS)	`sim/run_openems.py`
Grid near wires	1.0 mm	JUDG	Sensitivity runs at 0.5 / 1.0 / 2.0 mm	`sim/run_openems.py`
Grid in phantom	3.0 mm	JUDG	openEMS MRI tutorial uses 2 mm	`sim/run_openems.py`
Port / capacitor gap	2.0 mm	JUDG		`sim/run_openems.py`
Air spacer	200 mm	CONV	openEMS MRI_Loop_Coil tutorial	`sim/run_openems.py`
End criterion	1e-4	CONV	openEMS MRI_Loop_Coil tutorial	`sim/run_openems.py`
Excitation half-band	0.5 × f0	JUDG		`sim/run_openems.py`
Boundary	MUR (all sides)	CONV	openEMS MRI_Loop_Coil tutorial	`sim/run_openems.py`
Thin-wire effective radius a_eff	calibrated per grid on the R = 40 mm loop	DERIVED	Inverts Rosa & Grover eq. [62]; validated by V15b on the trefoil	`sim/cosim.py`
Copper conductivity	5.8e7 S/m	CONV	Standard handbook value, **not cited**	`sim/cosim.py`
Capacitor Q	1000	JUDG PLACEHOLDER	Until components are chosen	`sim/cosim.py`
Matching-inductor Q	100	JUDG PLACEHOLDER		`sim/cosim.py`
Preamp input impedance	2 Ω	JUDG PLACEHOLDER		`sim/cosim.py`
Tuning passes (arrays)	3	JUDG		`sim/cosim.py`

Parameter	Value	Tag	Rationale / source	Where
Tissue 7T (brain / skull / skin εr, σ)	52.7, 0.95 / 11.4, 0.29 / 38.9, 0.84	CONV (secondary)	openEMS bundled phantom, attributed to IT'IS/Gabriel; **not independently verified**	`tests/export_case.js` , app
Tissue 3T	62.8, 0.66 / 12.5, 0.13 / 46.8, 0.58	**APPROXIMATION**	openEMS 128 MHz *body* phantom values mapped to head layers	same
Phantom layers	skin 4 mm, skull 7 mm	JUDG PLACEHOLDER		`tests/export_case.js`
Test element	T(2,3), β 0.8, Rf 45 mm, 4 breaks, posterior midline at 20° elevation	JUDG		`tests/export_case.js`
Crossover target clearance (v1.9)	1.1 × (d_w + 2·t_i)	JUDG	10% margin over touching insulation, to absorb polyline sampling	`core/optimise.js` (O.crossovers)
Crossover taper width W (v1.9)	10 mm (pre-registered 6)	**TUNED**	Raised to keep the 2 mm bend limit on the pair cases (which V24 also tests); DECISIONS 2026-10-21	`core/optimise.js`
Crossover A_max, height step, cluster gap, r0_max (v1.9)	10 mm, 0.1 mm, 3 mm, 30 mm	JUDG	Search limits for bump height and plateau radius; reaching A_max is flagged (V24 d)	`core/optimise.js`
Grid refinement at crossovers REFINE (v1.9)	lines every 0.4 mm within ±3 mm	JUDG	Keeps two coils 1.65 mm apart several grid nodes apart (pre-run gate)	`sim/run_openems.py`
MIN_GAP / MIN_CELL (v1.9)	0.2 mm / 0.15 mm	JUDG	No sliver cells (they shrink the FDTD timestep); a case with a smaller cell aborts	`sim/run_openems.py`
COMSOL air sphere margin AIR (v1.10)	150 mm	JUDG	Distance from the farthest wire or tissue to the scattering boundary	`sim/comsol_build_mph.py`
COMSOL wire mesh size HMAX_WIRE (v1.10)	1 mm	JUDG	Same scale as the openEMS grid near wires; the PEC-edge inductance depends on it	`sim/comsol_build_mph.py`

Parameter	Value	Tag	Rationale / source	Where
			(calibrate on the loop, as for openEMS)	
COMSOL frequencies NF (v1.10)	41 over f0 (1 ± 0.4)	CONV (this project)	Same band as openEMS, fewer points (one solve per port)	`sim/comsol_build_mph.py`
Total thickness T (v1.8, added 2026-10-21)	0 mm default (= minimal lift, v1.7 geometry); app slider 0–60 mm	User input (Reza's design choice)	h = max(h_min, T/2); extra thickness goes outward only (Reza's choice). Slider range 0–60 mm is JUDG	`core/knot.js` (knot.height), app
Pair break shift (added 2026-10-21)	+0.100 × wire length (element 1), −0.100 (mirrored element 2)	JUDG rule, DERIVED value	Rule: largest worst-case port-gap-to-other-coil distance over all four pair cases, subject to break-to-crossing clearance ≥ 5 mm. Result: 7.3 mm (was 1.4 mm). Chosen on geometry only, before any pair result (DECISIONS 2026-10-21)	`tests/export_case.js`
Port terminal rounding PORT_ROUND (added 2026-10-21)	1 µm	DERIVED	Makes openEMS's 7-significant-digit XML export of the port box land exactly on its grid line	`sim/run_openems.py`
Timestep cap MAX_TS (added 2026-10-21)	2×10^6	JUDG	openEMS default is 10^9 (no real cap). A run that hits the cap is not trusted	`sim/run_openems.py`
V18 depth	along the ray from the head centre	JUDG	Not the true normal distance	`sim/cosim.py`

Design v1.11 (PROPOSED, pending Reza's approval): COMSOL link (`sim/comsol_run.py` CFG, `core/simcase.js`)

Parameter	Value	Tag	Rationale / source	Where
Gap length	2.0 mm	JUDG	Same as the openEMS model (GAP)	`sim/comsol_run.py`
Tube centreline	cubic spline through every case point, arc-length parameter	JUDG	Replaced a Fourier fit that missed bumped array wires by up to 9.8 mm (DECISIONS 2026-10-23)	`sim/comsol_run.py`
Tube patch length / NURBS knots	≤ 20 mm / 40 per direction	JUDG	COMSOL approximates each parametric patch with NURBS (its default is 20 knots)	`sim/comsol_run.py`
Air margin / PML thickness	60 / 80 mm	JUDG	Values of Will's `coils8_import.py`	`sim/comsol_run.py`
Mesh	physics-controlled, level 5 (normal) or 4 (fine)	JUDG	Will used 4; level 5 meshes a single trefoil in ~120k elements	`sim/comsol_run.py`, app
Match tolerance	|S11| ≤ −30 dB	JUDG	The handover's criterion for "matched"	`sim/comsol_run.py`
Stage-2 iterations / damping	10 / 0.7 (log space)	JUDG	Handover values (`tune_match_bounded`)	`sim/comsol_run.py`
Shunt Cp search range	0.5 pF – 2 nF	JUDG	Handover value	`sim/comsol_run.py`
Insulation permittivity εr	3	JUDG PLACEHOLDER	Typical enamel; real wire spec pending (D5). Editable in the app	`sim/comsol_run.py`, app
Crossing threshold for beads	3 mm centre to centre	JUDG	Catches every crossing (1.5–2 mm) and crossover	`sim/comsol_run.py`
Bead padding beyond the wire	0.6, then 2, then 3 mm	JUDG (mesh-driven)	The real 0.25 mm insulation meets the tube at a grazing angle the mesher cannot fill; 1.0 mm also failed on the trefoil (DECISIONS 2026-10-23)	`sim/comsol_run.py`

Parameter	Value	Tag	Rationale / source	Where
Touching-wire refusal	< 2·r_wire + 0.1 mm centre to centre	JUDG	Closer tubes would be fused by the geometry union (a short)	`sim/comsol_run.py`
Pre-flight: wrap stretch refusal	> 15%	JUDG	Coils that wrap cleanly stretch up to 12% (test arrays N 8–12, T 0–15 mm); the badly wrapped 94 mm coil stretched 27.5% (DECISIONS 2026-10-23)	`core/simcase.js`
Pre-flight: port or capacitor gap near another coil	< 1.5 mm centre to centre	JUDG	A 2 mm gap that close to another wire could short across it	`core/simcase.js` , `sim/comsol_run.py`
Pre-flight: wire point jump	longest step > 3 mm and > 5 × the median step	JUDG	Healthy case wires step 1.8–4.1 mm with no outliers; the badly wrapped coil jumped 56–62 mm	`sim/comsol_run.py`
Pre-flight: long-run warning	> 24 COMSOL solves	JUDG	About 25–65 s per solve; beyond 24 a run takes well over half an hour	`core/simcase.js`
Arrays: ports and beads	feed gap only, one port per coil; no insulation beads	JUDG (requester's choice; mesh-driven)	"only port is needed"; beads did not mesh on an 8-coil array and hardly change B1	`sim/comsol_run.py`
B1 map grid	60 × 60 per plane, the app's slice grid	JUDG	Same points as the app's quasi-static map	`sim/comsol_run.py` , app
B1 planes	axial / coronal / sagittal through the coils' centroid (coronal at 0.6 × centroid y)	JUDG	Puts the coronal plane inside the head for posterior coils	app
Single/pair position on the helmet	posterior midline, 20° elevation	JUDG	Same as the v1.7 cases (`tests/export_case.js`)	`core/simcase.js`
Pair gap shift search	120 shifts per gap period	JUDG	Maximises the gap-to-other-coil distance, gaps ≥ capacitor clearance from crossings	`core/simcase.js`

Parameter	Value	Tag	Rationale / source	Where
Default pair spacings	zero ± 0, 2, 4, 6 mm	JUDG	Brackets the Pair explorer’s zero	app
Comparison statistic	share of points within 20% after normalising, where the app map is within 30 dB of its maximum	JUDG	Information only (no criterion, by the requester’s choice)	app

Design v1.6 additions

Parameter	Value	Tag	Rationale / source	Where
Face zones: eyes	az ±17°, el +5°, half-widths 13° × 10°	JUDG PLACEHOLDER	Rough adult proportions. Replace with measured landmarks or the ICBM152 eye mask	`core/surface.js`
Face zone: mouth	az 0°, el −30°, half-widths 20° × 9°	JUDG PLACEHOLDER	Same	`core/surface.js`
Zone margin for wires	5 mm	JUDG PLACEHOLDER	Proposal default	`core/surface.js`
Lowest covered height zCut	−20 mm (was 0)	JUDG PLACEHOLDER	Lowered once zones protect the face	`core/surface.js`
Face wedge	off (was on)	JUDG	Replaced by zones	`core/surface.js`
lambdaZone	1 (was 0.05 in development)	JUDG	Wires entered zones at 0.05	`core/optimise.js`
Coverage buffer around zones	0.5 × footprint radius	JUDG	Stops the coverage term pulling wires into zones	`core/optimise.js`
Footprint zone check	97 points, 1.1 × Rf	JUDG	37 points at 1.0 × Rf missed zones; wrapped wires reach ~5% past Rf	`core/optimise.js`
Multi-start jitter	0.25 × spacing	JUDG		`core/optimise.js`
Adjacency tolerance	2 × sample spacing	DERIVED	Voronoi border detection	`core/optimise.js`
Ring count heuristic	round(√(N/2.5)) rings	JUDG		`core/optimise.js`
Ring elevation range	covered edge to 1.5 rad (start top at 1.3 rad)	JUDG		`core/optimise.js`
Ring search points per element	96	JUDG	Speed; final report uses 300	`core/optimise.js`
γ/2π (proton)	42.577478461 MHz/T	**LIT**	CODATA 2022 via NIST; VERIFIED 2026-10-20, archived in `sources/`	`core/knot.js`
Capacitor clearance from crossings	5 mm	JUDG PLACEHOLDER	Until the capacitor package size is known	app

Parameter	Value	Tag	Rationale / source	Where
Segment-length warning	λ/10 (free space)	JUDG	Rule of thumb, not a cited value	`core/knot.js`
Break placement	200 trial offsets; nudges up to ¼ segment	JUDG		`core/knot.js`

B1 quick look (Design v1.5)

Parameter	Value	Tag	Rationale / source	Where
Head surface	helmet semi-axes − 10 mm	JUDG PLACEHOLDER	Stand-in for a head; replace with a real head model when simulating	`core/field.js`
B1 units factor	100 µT/A per (1/mm)	DERIVED	µ0/4π with mm units; header of `core/field.js` ; checked by V8	`core/field.js`
Wire thinning for B1	≤ 200 points per element	JUDG	Speed. The exact finite-segment formula (since v1.6) is checked against Magpylib on the same thinned wires (V14, V22)	`core/field.js`
Colour scale	−30 to 0 dB relative to the maximum shown, viridis	CONV	Viridis is a standard perceptually uniform colour map	app
Array combination	$\sqrt{\Sigma\lvert B1_i\rvert^2}$ (1 A per element)	CONV	Common quick stand-in for combined array sensitivity; ignores noise correlation	app

Numerical choices

Parameter	Value	Tag	Rationale / check	Where
μ0/4π	1e-7 H/m = 0.1 nH/mm	DERIVED	SI definition	`core/inductance.js`
Near-field refinement distance	3 segment lengths	JUDG	Checked indirectly by V1, V2, V4	`core/inductance.js`
Max sub-pieces per segment	12	JUDG	Same	`core/inductance.js`
Kernel softening between elements	ε = d_w + 2 t_i	JUDG	Wires crossing between elements are separated by at least this. Sensitivity not yet tested	`core/optimise.js`
Self-inductance method	centreline vs copy offset by wire radius	LIT (method)	High-frequency surface-current formula, Rosa & Grover 1912 eq. [62], VERIFIED. Checked by V2	`core/inductance.js`
Element points (element view)	adaptive: 100·p·q (min 400), doubled until crossing count is stable	JUDG	Added after validation run 1	`core/knot.js`
Stage-1 table	128 pts; d = 0.3–3.0 Rf in 55 steps; 12 rotation steps per 2π/q	JUDG / DERIVED	Speed. Stage 2 corrects table error	`core/optimise.js`
Stage-2 / final points	160 / 300 per element	JUDG	V4: k converged to < 0.001 at 300	`core/optimise.js`
Self-L points	300 (solver); ≤ 800 (Element tab, thinned from the adaptive point count)	JUDG		`core/optimise.js`
Worst-pair weight λ_max	1 (cost term $\lambda_{max} \cdot k^4/thr^2$)	JUDG	Chosen before testing, not tuned. Made default by the pre-registered comparison, `validation/compare_worstpair.html`	`core/optimise.js`

Parameter	Value	Tag	Rationale / check	Where
Coverage weight λ	1	JUDG	Makes gaps dominate k² terms	`core/optimise.js`
Neighbour weight taper	2.0 → 2.4 Rf (footprint); 1.35 → 1.55 × null spacing (ring1)	JUDG	Stops pairs escaping by stepping just past the cut-off	`core/optimise.js`
Coverage samples	2400 on full ellipsoid	JUDG		`core/optimise.js`
Search sweeps	≤ 300 (stage 1), ≤ 12 (stage 2)	JUDG	Speed vs quality; not tuned	`core/optimise.js`
Random seed	1	JUDG	Reproducibility; stored in every export	`core/optimise.js`
Bloom floor formula	c / (R sin(π/p)) outward; c / (2 R sin(π/p)) symmetric	DERIVED	Claude's geometric argument; agrees with the numerical lift solver (see DECISIONS.md)	`core/knot.js`